

High-carbohydrate, high-fiber diets increase peripheral insulin sensitivity in healthy young and old adults.

To examine extra-alimentary effects of high-carbohydrate, high-fiber (HCF) diets, insulin-mediated glucose disposal employing the euglycemic clamp and hepatic glucose output (HGO) employing [6,6-2H₂]glucose were measured in 12 healthy young and old individuals before and after 21-28 d of an HCF diet. Diet lowered fasting concentrations of glucose from 5.3 $\pm$ 0.2 to 5.1 $\pm$ 0.1 mmol/L (p less than 0.01) and insulin from 66.0 $\pm$ 7.9 to 49.5 $\pm$ 5.7 pmol/L (p less than 0.01). Fasting serum cholesterol decreased from 5.17 $\pm$ 0.18 to 3.80 $\pm$ 0.20 mmol/L (p less than 0.01) in young individuals and from 6.15 $\pm$ 0.52 to 4.99 $\pm$ 0.49 mmol/L (p less than 0.01) in elderly individuals. Fasting serum triglyceride concentrations, basal HGO, and insulin suppression of HGO were unchanged by the diet. Glucose disposal rates increased from 18.87 $\pm$ 1.66 before 23.87 $\pm$ 2.78 μ mol.kg⁻¹.min⁻¹ after the diet (p less than 0.02). Therefore, HCF diets may improve carbohydrate economy by enhanced peripheral sensitivity to insulin.